

Start Here Guide

For

A-GRA

5.0a

21 APR 2026

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Revision History

Version	Description	Generation Date
5.0a	Initial Public ASK 5.0a Release	21 APR 2026

Mission Autonomy Introduction

Mission Autonomy (MA) is the name of the software solution Autonomous Collaborative Platforms (ACPs) require to manage the mission-level execution, planning, prioritization, and interaction with other systems that humans manage in crewed platforms. MA interfaces are standardized to support rapid capability development while enabling interoperability with MA peers and human controlled systems. MA solutions are sourced via competitive selection of best-in-breed capabilities.

This package defines the Autonomy Government Reference Architecture (A-GRA) which captures the MA interface standards and supporting documentation. Adherence to this architecture enables implementers to develop MA solutions that can be more easily integrated into existing platforms, quickly upgraded, and ported between modular compatible MA solutions. It also enables integrators to rapidly deploy custom, performant MA solutions to accomplish diverse and evolving mission use cases.

Operational View-1

This Operational View-1 (OV-1) figure depicts a high-level notional mission use case for a Defensive Counter Air (DCA) mission. ACPs take-off from an airbase under oversight from a Launch and Recovery Element (LRE) and fly a hold pattern where they await the arrival of a Quarterback (QB). The ACPs are assigned to a team and tasked with transit in formation to the Mission Area. Once they arrive, ACPs allocate coverage of the zone and fly Combat Air Patrol (CAP) patterns. When a threat emerges, MA forwards a request to engage the threat to the QB. The QB can simply approve the MA generated strike request while maintaining a safe standoff distance. The ACP team will self-nominate a shooter, engage the threat, and return to CAP.

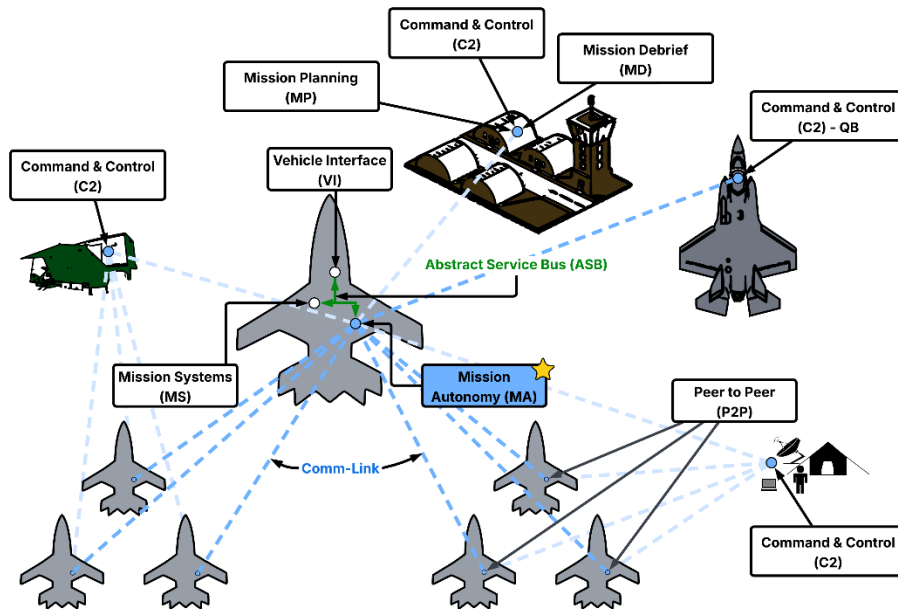


A high level OV-1 used

to guide interface development for the A-GRA

Potential alternate QBs are depicted around the periphery of the battlespace and may interact with ACPs according to permissions associated with their roles. When an ACP reaches the Bingo Fuel state, it requests Return to Base (RTB) to the primary or alternate launch and recovery site. The sense-and-avoid traffic deconfliction maneuver depicted as part of the primary RTB route illustrates the robustness of the ACP to uncertainties in the operational environment. Finally, the ACP lands at the designated airfield.

L1 Interface Diagram



and its six L1 interfaces including four distinct C2 nodes

The scalability and adaptability required of MA solutions to accomplish complex, large-scale mission operations are enabled by rigorous definition of the interface specifications between MA and the other systems in the operational vignette. This ecosystem is broken down into multiple architecture levels to track the interface requirements for the MA and other systems. Level 1 (L1) is the system level, which defines the external interfaces of the MA to the other systems. The L1 interfaces are Command and Control (C2), Vehicle Interface (VI), Mission Systems (MS), Mission Planning (MP), Mission Debrief (MD) and Peer to Peer (P2P). All of these interfaces are described in detail in the A-GRA Interface Control Document (ICD).

Level 2 (L2) is the component level, which defines the interfaces within MA. MA L2 interface definitions will be included in future releases of the A-GRA.

A-GRA Organization

The A-GRA provides the necessary definitions and requirements by which autonomy components and systems may be developed but does not exhaustively detail all specific implementation options. This is a high level description of interactions organized by interface:

- C2
 - Describes three categories of C2 interactions – Direct, Indirect, and Responsive
 - Defines interactions to form and task a team
 - Defines Role Based Access Control (RBAC) for ACPs
 - Describes command flows for employing weapons
- Vehicle Interface (VI)
 - Describes the responsibilities of MA and Flight Autonomy (FA) with regard to safety-criticality and management of the platform data
 - Identifies control mode messages and sequences for communication with FA. Control modes include Heading, Speed, Altitude (HSA), Waypoint Following, and Curve Following
 - Defines the required subsystem status reporting and vehicle state data to be provided by FA to MA
 - Defines an interface for FA to communicate command validation (accept/reject) back to MA. FA always retains control but 'listens' to MA based on mission state
- Mission Systems (MS) Interface
 - Exchange of messages between A-GRA compliant platforms. Includes communications systems such as the Decentralized Message Service (DMS)
 - Providing status data to MA such as Position, Navigation, and Timing (PNT) data and system status of the local and remote platforms
 - Provides interfaces for sensor and comms system interactions
 - Defines interfaces for Weapons related interactions
- Mission Planning (MP) Interface
 - Defines the Mission Data Package which many MA interactions depend upon being loaded pre-mission
 - Defines a system of plans and sub-plans
 - Defines the Rules of Engagement (ROE) settings and other crucial configuration settings for ACPs
- Mission Debrief (MD) Interface
 - Defines the use cases for mission debrief and replay of MA data
 - Defines requirements for on-board debrief systems and support for in-flight mission debrief
- Peer-to-Peer (P2P) Interface
 - Defines the process for team formation and leader election
 - Defines interactions to distribute the package Common Operating Picture (COP)
 - Defines peer contingencies and state data sharing requirements

A-GRA interactions are built with messages from the Universal Command and Control Interface (UCI) plus extensions for MA. The source schema for Autonomy implementors includes all of

the base UCI messages and MA extensions in the form of Extensible Markup Language (XML) Schema Definition (XSD) files.

Compliance with the A-GRA is currently based on properly implementing interactions for desired outcomes. Interactions, or sequences, must use the correct messages and populate required fields with valid data. An automated compliance test harness is available for MA developers to evaluate L1 interface compliance. It currently performs sequence/message level validation and will be enhanced in future releases of the A-GRA.

Acronyms and Abbreviations

Abbreviation	Definition
ACP	Autonomous Collaborative Platform
A-GRA	Autonomy Government Reference Architecture
ASB	Abstract Service Bus
C2	Command and Control
CAP	Combat Air Patrol
COP	Common Operating Picture
DCA	Defensive Counter Air
FA	Flight Autonomy
HSA	Heading, Speed, Altitude
ICD	Interface Control Document
L1	Level 1 (MA ↔ External Interface)
L2	Level 2 (Intra MA Interface)
LRE	Launch and Recovery Element
MA	Mission Autonomy
MD	Mission Debrief
MP	Mission Planner
MS	Mission System(s)
OV-1	High Level Operational Concept Operational View
P2P	Peer-to-Peer

PNT	Position, Navigation, Timing
QB	Quarterback
ROE	Rules of Engagement
RTB	Return to Base
UCI	Universal Command and Control Interface
VI	Vehicle Interface
WEZ	Weapon(s) Engagement Zone
XML	Extensible Markup Language
XSD	XML Schema Definition

Contact Information

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